

SEYED AMIR KASAEI

Research Assistant, RIML Lab & M.Sc. Student, Computer Engineering, Sharif University of Technology

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EDUCATION

- **M.Sc. in Computer Engineering**
Sharif University of Technology - (Sep 2023 – Present)
Focus: *Compositional Generation in T2I Diffusion Models*
Supervisor: [Dr. M H Rohban](#)
GPA: 18.54/20
- **B.Sc. in Computer Engineering**
University of Guilan (Sep 2019 – Aug 2023)
Focus: *Medical Image Analysis*
Supervisor: [Dr. S A Mirroshandel](#)
GPA: 19.58/20 - Ranked 1st

RESEARCH INTERESTS

- Computer Vision, Compositional and Multimodal Learning across Vision and Language
- Text-to-Image Generation and Diffusion Models
- Visual Reasoning and Semantic Alignment in Cross-Modal Systems

PUBLICATIONS

- **CARINOX: Inference-time Scaling with Category-Aware Reward-based Initial Noise Optimization and Exploration**
Seyed Amir Kasaei, Ali Aghayari, Arash Marioriyad, Niki Sepasian, Shayan Baghayi Nejad, MohammadAmin Fazli, Mahdieh Soleymani Baghshah, Mohammad Hossein Rohban
TMLR 2026
arxiv.org/abs/2509.17458
- **Evaluating the Evaluators: Metrics for Compositional Text-to-Image Generation**
Seyed Amir Kasaei, Ali Aghayari, Arash Marioriyad, Niki Sepasian, MohammadAmin Fazli, Mahdieh Soleymani Baghshah, Mohammad Hossein Rohban
NeurIPS 2025 Workshop GenProCC
arxiv.org/abs/2509.21227
- **Hidden Meanings in Plain Sight: RebusBench for Evaluating Cognitive Visual Reasoning**
Seyed Amir Kasaei, Arash Marioriyad, Mahbod Khaleti, MohammadAmin Fazli, Mahdieh Soleymani Baghshah, Mohammad Hossein Rohban
ICLR 2026 Workshop: From Human Cognition to AI Reasoning (HCAIR)
arxiv.org/abs/2604.01764
- **Erasure or Erosion? Evaluating Compositional Degradation in Unlearned Text-To-Image Diffusion Models**
Arian Komaei Koma, Seyed Amir Kasaei, Ali Aghayari, AmirMahdi Sadeghzadeh, Mohammad Hossein Rohban
CVPR 2026 Workshop on Machine Unlearning for Computer Vision
arxiv.org/abs/2604.04575
- **Weeding Out Bad Seeds: Noise-Robust Unlearning for Text-to-Image Diffusion Models**
Arian Komaei Koma, Seyed Amir Kasaei, Ali Aghayari, Aida Aryafar, Matin Ghiasi, Amirhossein Sour, Mohammad Mosayyebi, AmirMahdi Sadeghzadeh, Mohammad Hossein Rohban
Under Review at NeurIPS 2026
- **Erased but Exploitable: Black-box Embedding-Aware Prompting Against Unlearned Text-to-Image Diffusion Models**
Arian Komaei Koma, Seyed Amir Kasaei, AmirMahdi Sadeghzadeh, Mohammad Hossein Rohban
Under Review at TMLR
arxiv.org/abs/2605.26332

EXPERIENCE

Research Assistantships

- *RIML Lab, Sharif University of Technology*
Supervisor: [Dr. Mohammad Hossein Rohban](#)
Focus: Compositional Generation, T2I Diffusion Models, Visual Reasoning, Multimodal Models
Sep 2023 – Present

Teaching Assistantships (Sharif University of Technology)

- *Data Mining* (Head TA) — Instructor: [Dr. Mohammad Amin Fazli](#) Sep 2025 – Jan 2026
Duties: Led course coordination, quiz design, and grading; assisted with lectures, assignments, and student support.
- *System 2 in AI* — Instructors: [Dr. Mohammad Hossein Rohban](#), [Dr. Mahdieh Soleymani Baghshah](#) Feb 2025 – Jun 2025
Topics: Compositional Generation, Inference-time scaling, Test-time adaptation | *Duties:* Material prep and grading.
- *Deep Learning* — Instructor: [Dr. Mahdieh Soleymani Baghshah](#) Feb 2025 – Jun 2025
Topics: Generative Models, VAE, GAN, Diffusion | *Duties:* Quiz and exam design, and grading.
- *Intelligent Analysis of Biomedical Images* — Instructor: [Dr. Mohammad Hossein Rohban](#) Sep 2024 – Jan 2025
Topics: Diffusion and Generative Models | *Duties:* Supported exercises, exam/quiz design, and grading.

NOTABLE SELECTED COURSES

Master of Science

- Data Science in Software Engineering - 19 / 20
- Intelligent Analysis of Biomedical Images - 20 / 20
- Deep Learning - 16.7 / 20
- Algorithmic Game Theory - 19.5 / 20

Bachelor of Science

- Fundamental of Data Mining - 20 / 20
- Introduction to Deep Learning - 19.8 / 20
- Introduction to Machine Learning - 18.16 / 20
- Fundamental of Language and Speech Processing - 20 / 20

HONORS & AWARDS

- **Exceptional Talent Admission**, Sharif University of Technology Sep 2023
Awarded direct admission to M.Sc. in Computer Engineering as an exceptional talented student.
- **Ranked 1st**, B.Sc. Computer Engineering, University of Guilan 2019 – 2023
Highest GPA among all undergraduate Computer Engineering students.

SKILLS

- **Machine Learning & Computer Vision:** Generative Models, Multi-modals, Diffusion, VLMs, Visual Reasoning
- **Programming Languages & Frameworks:** Python, PyTorch, TensorFlow, Keras, Java
- **Data Science & Developer Tools:** NumPy, Pandas, Matplotlib, W&B, Git
- **Soft Skills & Academic Leadership:** Teamwork, Leadership, Technical Communication, Mentorship, Problem Solving, Collaboration, Time Management

LICENSES & CERTIFICATIONS

- **IELTS** - [Certificate](#)
IDP - Dec 2025, Overall: 7.5, Listening: 9, Reading: 7.0, Writing: 6.5, Speaking: 7.0

REFERENCES

Dr. Mohammad Hossein Rohban rohban@sharif.edu Associate Professor of Computer Engineering, Sharif University of Technology, Tehran, Iran	<i>Sharif University of Technology</i> Google Scholar
Dr. Mohammad Amin Fazli fazli@sharif.edu Associate Professor of Computer Engineering, Sharif University of Technology, Tehran, Iran	<i>Sharif University of Technology</i> Google Scholar
Dr. Mahdieh Soleymani Baghshah soleymani@sharif.edu Associate Professor of Computer Engineering, Sharif University of Technology, Tehran, Iran	<i>Sharif University of Technology</i> Google Scholar